Autocorrelation of the measured input gain and the augmented estimator

2026-10-13 z axis; lengths in nm, motion gains divided by a_{nom}

1 The question

06-13 run (6-state estimator, $y_2 = \bar{a}_m$): 1 s descent to 1.2 R, then 1 Hz oscillation 1.2–3.42 R; $\lambda_c = 0.7$, $a_{\mathrm{pd}} = a_{\mathrm{cov}} = 0.05$; seed 1

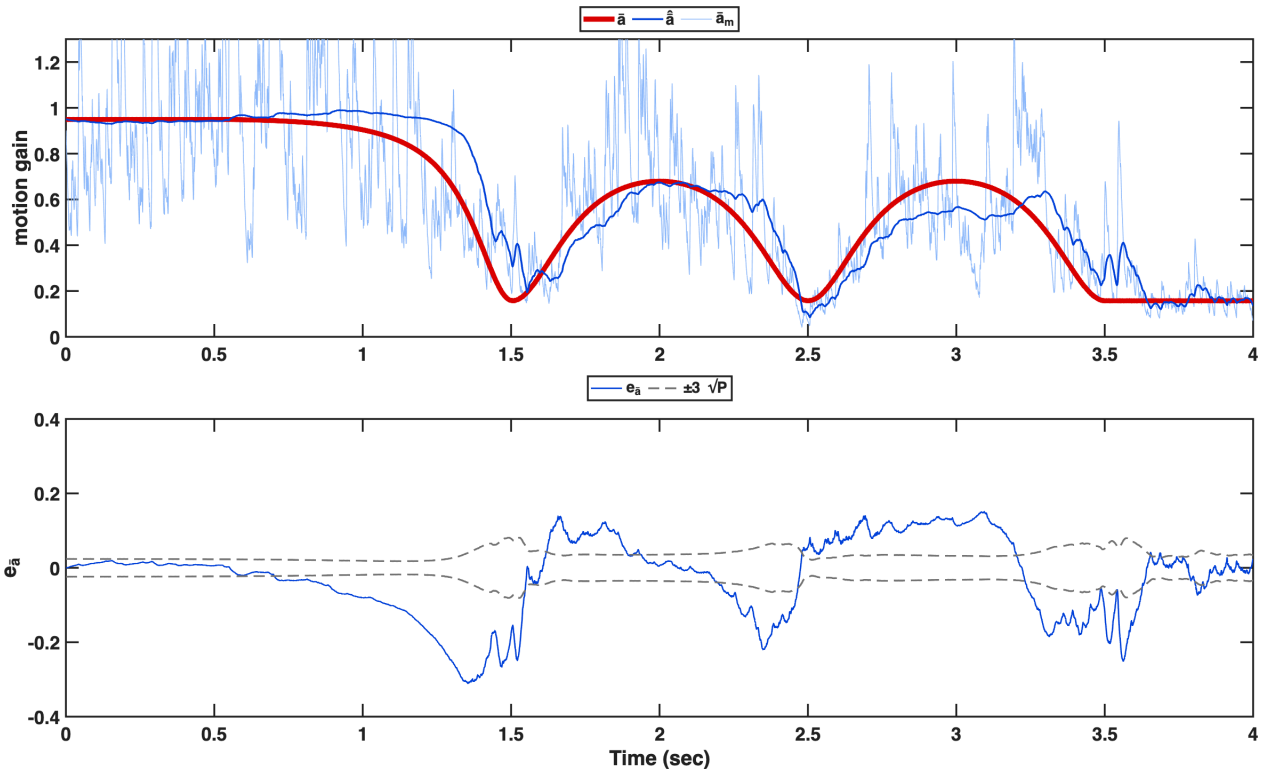
$$\overline{\delta\bar{x}_m}[k+1] = a_{\mathrm{pd}}\,\delta\bar{x}_m[k] + (1 - a_{\mathrm{pd}})\,\overline{\delta\bar{x}_m}[k]$$

$$\delta\bar{x}_r[k] = \delta\bar{x}_m[k] - \overline{\delta\bar{x}_m}[k]$$

$$\hat{\sigma}^2_{\delta\bar{x}_r}[k+1] = a_{\mathrm{cov}}\,\delta\bar{x}_r^2[k] + (1 - a_{\mathrm{cov}})\,\hat{\sigma}^2_{\delta\bar{x}_r}[k]$$

$$\bar{a}_m[k] = \frac{\hat{\sigma}^2_{\delta\bar{x}_r}[k] - C_n\,\sigma_n^2}{C_T\,\kappa_T}$$

$$y_2[k] = \bar{a}_m[k], \qquad r_2 = \sigma^2_{\bar{a}_m}$$

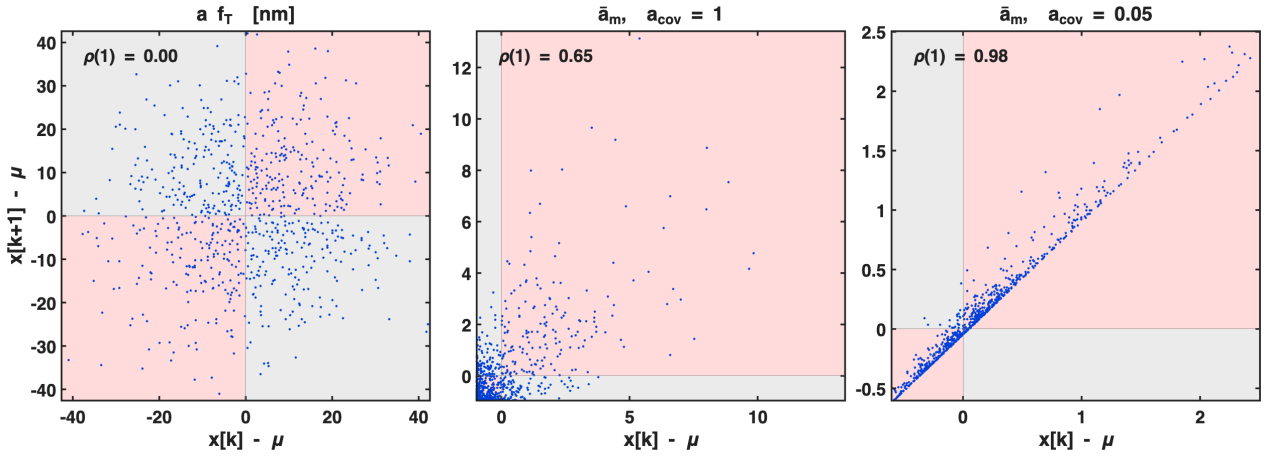


2 Autocorrelation

record: far field 0.25–0.75 s, $n = 800$ samples; τ: steps between the two samples. Figure: $\tau = 1$, each point one pair $(x[k] - \mu,\, x[k+1] - \mu)$, $n - 1$ points of seed 1; red quadrants: product > 0, grey: product < 0; $\rho(1)$: 200 seeds

$$\rho(\tau) = \frac{\dfrac{1}{n-\tau}\displaystyle\sum_{k=1}^{n-\tau} \big(x[k] - \mu\big)\big(x[k+\tau] - \mu\big)}{\dfrac{1}{n}\displaystyle\sum_{k=1}^{n} \big(x[k] - \mu\big)^2}$$

$$x[k+\tau] - \mu \approx \rho(\tau)\,\big(x[k] - \mu\big)$$



3 What the estimator assumes

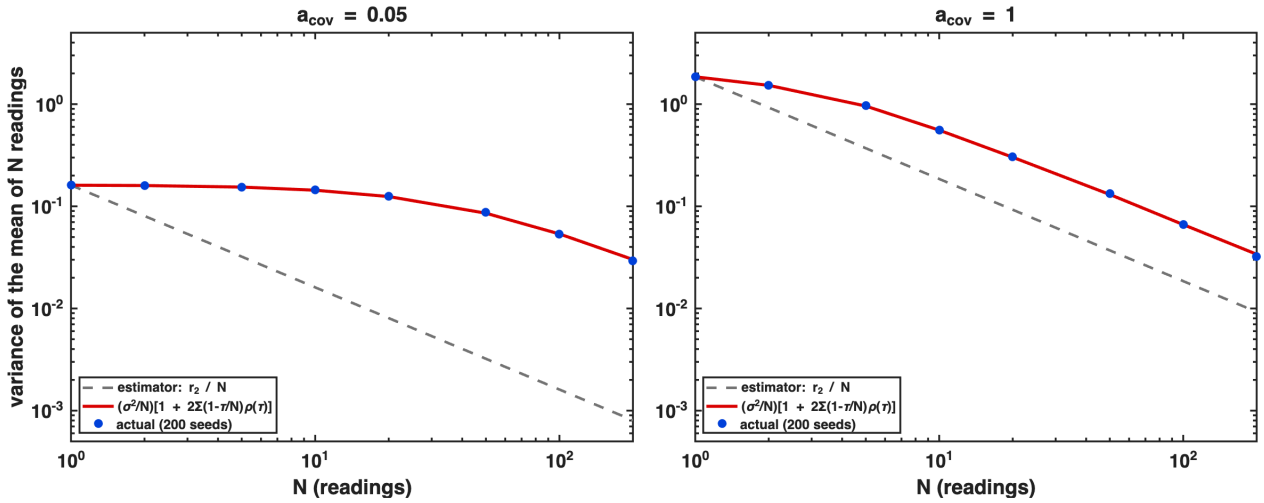
same record (motion gain constant within 1%); N consecutive readings averaged. Dots: each run cut into blocks of N, variance of the block means over 200 seeds; grey: $\sigma_{y_2}^2/N$ (the estimator with $r_2 = \sigma_{y_2}^2$); red: the second line with $\rho(\tau)$ of page 2; at $N = 1$ all three are $\sigma_{y_2}^2$

$$\frac{1}{p_{44}[k+1]} = \frac{1}{p_{44}[k]} + \frac{1}{r_2}$$

$$p_{44}[N] = \frac{r_2}{N}$$

$$\sigma^2\Big(\frac{y_2[1]+y_2[2]}{2}\Big) = \frac{\sigma_{y_2}^2}{2}\big[1+\rho(1)\big]$$

$$\sigma^2\Big(\frac{1}{N}\sum_{k=1}^{N} y_2[k]\Big) = \frac{\sigma_{y_2}^2}{N}\Big[1+2\sum_{\tau=1}^{N-1}\Big(1-\frac{\tau}{N}\Big)\rho(\tau)\Big]$$



4 Why it matters: the process noise of the motion gain

first-order estimator of Section 4 of motion_ctrl: 22.2 → 1.10 R in 1 s, then triangle 1.10 ↔ 10 R at 2 Hz; plane wall, $\lambda_c = 0.4$, seed 1; $a_{\mathrm{cov}} = 0.05$ with $\varepsilon = 10^{-3}$ diverges (not drawn)

$$q_{44} = \varepsilon\, r_2$$

$$a_{\mathrm{cov}} = 0.05: \quad r_2 = \sigma_{y_2}^2$$

$$a_{\mathrm{cov}} = 1: \quad r_2 = \mathrm{IF} \cdot \sigma_{y_2}^2, \qquad \mathrm{IF} = 1 + 2\sum_{\tau \geq 1} \rho_{y_2}(\tau)$$

